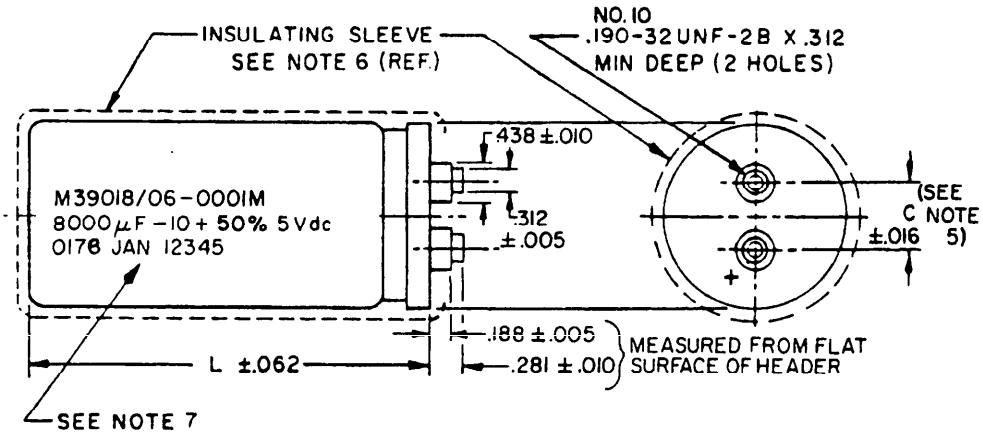
MILITARY SPECIFICATION SHEET

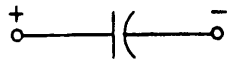
CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE) (POLARIZED), ESTABLISHED RELIABILITY, STYLE CUR91 (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39018.



INCHES	MM	INCHES	MM
.005	.13	.500	12.70
.010	.25	.875	22.22
.016	.41	1.125	28.58
.062	1.57	1.250	31.75
.188	4.78	1.375	34.92
.281	7.14	2.000	50.80
.312	7.92	2.500	63.50
.438	11.13	3.000	76.20



SEE NOTE 4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Capacitors are mounted by means of a wraparound, footed-type bracket.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. "C" dimension is .500 for case diameter of 1.375; .875 for case diameter of 2.000; 1.125 for case diameter of 2.500; and 1.250 for case diameter of 3.000.
6. Insulating sleeve thickness shall not exceed .016 and shall overlap the ends of the capacitor body.
7. Example of marking and its relative location on the capacitor body.

FIGURE 1. Style CUR91.

FSC 5910

Ⓐ denotes changes

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CUR91 Case Code

Case size	Bare case dimensions	
	D	L .062
AA	1.375	2.125
AB	1.375	3.125
AC	1.375	4.125
AD	1.375	4.625
AE	1.375	5.125
AF	1.375	5.625
BB	2.000	3.125
BC	2.000	4.125
BD	2.000	4.625
BE	2.000	5.125
BF	2.000	5.625
CB	2.500	3.125
CC	2.500	4.125
CD	2.500	4.625
CE	2.500	5.125
CF	2.500	5.625
DC	3.000	4.125
DD	3.000	4.625
DE	3.000	5.125
DF	3.000	5.625

INCHES	MM
.093	2.36
1.375	34.92
2.000	50.80
2.125	53.98
2.500	63.50
3.000	76.20
3.125	79.38
4.125	104.78
4.625	117.48
5.125	130.18
5.625	142.88

FIGURE 1. Style CUR91 capacitors - Continued.

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TABLE I. Style CUR91 capacitors.

DC rated voltage @ 105°C	Capacitance (nom)	DC surge voltage @ 105°C	ESR (max) Ⓐ	Impedance (max) @ -55°C (120 Hz)	DC leakage (max)		Maximum AC ripple current 1/ 120 Hz @ 85°C	Case size	Dash number 2/
					25°C	105°C			
Volts	µF	Volts	Ohms	Ohms	mA		Amps		
5	8,000	7	.045	1.28	0.30	2.70	6.0	AA	0001-
5	17,000	7	.029	0.61	0.44	3.96	8.0	AB	0002-
5	25,000	7	.023	0.41	0.53	4.77	9.2	AC	0003-
5	29,000	7	.018	0.36	0.57	5.13	10.4	AD	0004-
5	34,000	7	.015	0.31	0.62	5.58	11.6	AE	0005-
5	37,000	7	.013	0.28	0.65	5.85	12.2	AF	0006-
5	40,000	7	.019	0.28	0.67	6.03	12.6	BB	0007-
5	60,000	7	.014	0.19	0.82	7.38	15.1	BC	0008-
5	67,000	7	.013	0.17	0.87	7.83	16.3	CB	0009-
5	70,000	7	.012	0.16	0.89	8.01	16.3	BD	0010-
5	80,000	7	.012	0.14	0.95	8.55	16.6	BE	0011-
5	90,000	7	.010	0.13	1.00	9.00	16.9	BF	0012-
5	100,000	7	.010	0.12	1.06	9.54	19.1	CC	0013-
5	110,000	7	.010	0.11	1.11	9.99	21.5	CD	0014-
5	130,000	7	.009	0.10	1.21	10.90	23.0	CE	0015-
5	150,000	7	.008	.091	1.30	11.70	23.8	CF	0016-
5	150,000	7	.010	.091	1.30	11.70	24.1	DC	0017-
5	170,000	7	.010	.091	1.38	12.40	24.7	DD	0018-
5	200,000	7	.009	.084	1.50	13.50	26.6	DE	0019-
5	220,000	7	.008	.084	1.57	14.10	26.8	DF	0020-
10	5,000	15	.045	0.84	0.34	3.06	6.0	AA	0021-
10	10,000	15	.028	0.42	0.47	4.23	8.2	AB	0022-
10	15,000	15	.022	0.28	0.58	5.22	9.4	AC	0023-
10	17,000	15	.018	0.25	0.62	5.58	10.4	AD	0024-
10	20,000	15	.015	0.21	0.67	6.03	11.4	AE	0025-
10	22,000	15	.015	0.19	0.70	6.30	12.2	AF	0026-
10	24,000	15	.020	0.18	0.73	6.57	12.2	BB	0027-
10	37,000	15	.016	0.12	0.91	8.21	14.6	BC	0028-
10	42,000	15	.016	0.11	0.97	8.75	16.0	CB	0029-
10	43,000	15	.013	0.10	0.98	8.85	15.4	BD	0030-
10	48,000	15	.012	.093	1.04	9.35	16.3	BE	0031-
10	55,000	15	.011	.081	1.11	9.99	16.8	BF	0032-
10	64,000	15	.011	.069	1.20	10.8	19.0	CC	0033-
10	75,000	15	.010	.060	1.30	11.7	21.5	CD	0034-
10	86,000	15	.009	.054	1.39	12.5	22.2	CE	0035-
10	94,000	15	.014	.051	1.45	13.1	20.0	DC	0036-
10	96,000	15	.010	.051	1.47	13.2	23.0	CF	0037-
10	110,000	15	.010	.048	1.57	14.1	22.7	DD	0038-
10	120,000	15	.010	.045	1.64	14.8	23.4	DE	0039-
10	140,000	15	.010	.042	1.77	16.0	24.2	DF	0040-
15	4,000	20	.043	1.04	0.37	3.31	6.5	AA	0041-
15	8,000	20	.025	0.52	0.52	4.68	8.7	AB	0042-
15	12,000	20	.024	0.35	0.64	5.73	9.0	AC	0043-
15	14,000	20	.019	0.30	0.69	6.19	10.7	AD	0044-
15	16,000	20	.017	0.26	0.74	6.61	12.0	AE	0045-
15	18,000	20	.016	0.23	0.78	7.01	13.6	AF	0046-

See footnotes at end of table.

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TABLE I. Style CUR91 capacitors - Continued.

DC rated voltage @ 105°C	Capacitance	DC surge voltage @ 105°C	ESR (max) Ⓐ	Impedance (max) @ -55°C (120 Hz)	DC leakage (max)		Maximum AC ripple current 1/ 120 Hz @ 85°C	Case size	Dash number 2/
					85°C	105°C			
Volts	µF	Volts	Ohms	Ohms	mA		Amps		
15	20,000	20	.018	0.22	0.82	7.39	13.0	BB	0047-
15	30,000	20	014	0.14	1.00	9.05	15.2	BC	0048-
15	35,000	20	.011	0.12	1.09	9.78	17.1	BD	0049-
15	40,000	20	.010	0.11	1.16	10.5	18.2	BE	0050-
15	42,000	20	.012	0.11	1.19	10.7	18.2	CB	0051-
15	45,000	20	.010	.096	1.23	11.1	19.5	BF	0052-
15	51,000	20	.010	.087	1.31	11.8	20.7	CC	0053-
15	59,000	20	.010	.075	1.41	12.7	21.8	CD	0054-
15	68,000	20	.009	.066	1.51	13.6	23.1	CE	0055-
15	76,000	20	.008	.060	1.60	14.4	24.5	CF	0056-
15	76,000	20	.010	.060	1.60	14.4	23.1	DC	0057-
15	89,000	20	.009	.054	1.73	15.6	25.7	DD	0058-
15	100,000	20	.009	.051	1.84	165.	26.6	DE	0059-
15	110,000	20	.008	.048	1.93	17.3	27.5	DF	0060-
20	2,800	30	.044	1.47	0.35	3.19	6.1	AA	0061-
20	5,100	30	.026	0.81	0.48	4.31	8.5	AB	0062-
20	8,600	30	.020	0.48	0.62	5.60	9.9	AC	0063-
20	10,000	30	.019	0.41	0.67	6.04	11.0	AD	0064-
20	11,000	30	.017	0.37	0.70	6.33	11.7	AE	0065-
20	13,000	30	.016	0.32	0.76	6.88	12.0	AF	0066-
20	20,000	30	.013	0.21	0.95	8.54	15.4	BC	0067-
20	23,000	30	.013	0.18	1.02	9.16	15.8	BD	0068-
20	24,000	30	.016	0.17	1.04	9.35	16.0	CB	0069-
20	26.000	30	.012	0.16	1.08	9.73	16.5	BE	0070-
20	30,000	30	.012	0.14	1.16	10.5	17.2	BF	0071-
20	37,000	30	.012	0.11	1.29	11.6	18.8	CC	0072-
20	43,000	30	.011	0.10	1.39	12.5	19.9	CD	0073-
20	47,000	30	.010	.090	1.45	13.1	21.2	CE	0074-
20	54,000	30	.012	.081	1.56	14.0	22.0	DC	0075-
20	55,000	30	.009	.078	1.57	14.2	22.6	CF	0076-
20	63,000	30	.009	.069	1.68	15.2	24.5	DD	0077-
20	73,000	30	.009	.063	1.81	16.3	25.9	DE	0078-
20	82,000	30	.008	.057	1.92	17.3	26.5	DF	0079-
25	2,500	40	.050	1.64	0.38	3.38	5.7	AA	0080-
25	5,500	40	.031	0.75	0.56	5.00	7.7	AB	0081-
25	7,500	40	.021	0.55	0.65	5.85	9.6	AC	0082-
25	8,700	40	.020	0.47	0.70	6.30	10.0	AD	0083-
25	10,000	40	.019	0.41	0.75	6.75	11.0	AE	0084-
25	11,000	40	.018	0.37	0.79	7.08	12.0	AF	0085-
25	17,000	40	.015	0.25	0.98	8.80	14.5	BC	0086-
25	20,000	40	.012	0.21	1.06	9.54	16.1	BD	0087-
25	20,000	40	.011	0.21	1.06	9.54	16.8	CB	0088-
25	23,000	40	.011	0.18	1.14	10.2	16.8	BE	0089-
25	25,000	40	.011	0.17	1.19	10.7	17.4	BF	0090-
25	30,000	40	.011	0.14	1.30	11.7	19.6	CC	0091-
25	35,000	40	.010	0.12	1.40	12.6	20.7	CD	0092-
25	40,000	40	.009	0.11	1.50	13.5	22.1	CE	0093-
25	44,000	40	.015	0.10	1.57	14.2	19.4	DC	0094-

See footnotes at end of table.

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TABLE I. Style CUR91 capacitors - Continued.

DC rated voltage @ 105°C	Capacitance	DC surge voltage @ 105°C	ESR (max) Ⓐ	Impedance (max) @ -55°C (120 Hz)	DC leakage (max)		Maximum AC ripple current 1/ 120 Hz @ 85°C	Case size	Dash number 2/
					25°C	105°C			
Volts	µF	Volts	Ohms	Ohms	mA		Amps		
25	45,000	40	.009	.093	1.59	14.3	23.2	CF	0095-
25	52,000	40	.009	.081	1.71	15.4	25.5	DD	0096-
25	60,000	40	.009	.072	1.84	16.5	26.5	DE	0097-
25	67,000	40	.008	.066	1.94	17.5	27.5	DF	0098-
30	2,200	45	.054	1.86	0.39	3.47	5.5	AA	0099-
30	4,400	45	.033	0.93	0.54	4.90	7.5	AB	0100-
30	6,600	45	.025	0.62	0.67	6.00	8.8	AC	0101-
30	7,700	45	.020	0.53	0.72	6.49	9.7	AD	0102-
30	8,800	45	.019	0.47	0.77	6.94	11.5	AE	0103-
30	9,900	45	.017	0.41	0.82	7.36	11.7	AF	0104-
30	10,000	45	.020	0.41	0.82	7.39	12.0	BB	0105-
30	15,000	45	.015	0.28	1.00	9.06	14.3	BC	0106-
30	17,000	45	.014	0.24	1.07	9.64	15.1	BD	0107-
30	18,000	45	.016	0.23	1.10	9.92	15.8	CB	0108-
30	20,000	45	.012	0.21	1.16	10.5	15.9	BE	0109-
30	22,000	45	.012	0.19	1.22	11.0	17.0	BF	0110-
30	26,000	45	.012	0.16	1.32	11.9	18.6	CC	0111-
30	30,000	45	.010	0.14	1.42	12.8	20.3	CD	0112-
30	34,000	45	.010	0.12	1.51	13.6	21.0	CE	0113-
30	38,000	45	.015	0.11	1.60	14.4	19.2	DC	0114-
30	39,000	45	.010	0.11	1.62	14.6	21.8	CF	0115-
30	45,000	45	.010	.093	1.74	15.7	23.3	DD	0116-
30	52,000	45	.009	.081	1.87	16.9	25.3	DE	0117-
30	58,000	45	.008	.075	1.98	17.8	27.0	DF	0118-
40	1,800	55	.065	2.28	0.40	3.62	5.0	AA	0119-
40	3,700	55	.038	1.11	0.58	5.19	8.0	AB	0120-
40	5,500	55	.033	0.75	0.70	6.33	9.2	AC	0121-
40	6,400	55	.023	0.64	0.76	6.83	10.7	AD	0122-
40	7,400	55	.020	0.56	0.82	7.34	11.6	AE	0123-
40	8,200	55	.018	0.50	0.86	7.73	12.4	AF	0124-
40	9,000	55	.019	0.46	0.90	8.10	12.4	BB	0125-
40	13,000	55	.016	0.32	1.08	9.73	14.7	BC	0126-
40	15,000	55	.014	0.28	1.16	10.5	16.0	BD	0127-
40	15,000	55	.014	0.28	1.16	10.5	16.8	CB	0128-
40	18,000	55	.013	0.23	1.27	11.5	17.9	BE	0129-
40	19,000	55	.011	0.22	1.31	11.8	18.1	BF	0130-
40	22,000	55	.011	0.19	1.41	12.7	19.6	CC	0131-
40	25,000	55	.010	0.17	1.50	13.5	21.0	CD	0132-
40	28,000	55	.009	0.15	1.59	14.3	22.3	CE	0133-
40	32,000	55	.009	0.13	1.70	15.3	22.2	DC	0134-
40	33,000	55	.010	0.13	1.72	15.5	23.1	CF	0135-
40	37,000	55	.010	0.11	1.82	16.4	24.2	DD	0136-
40	43,000	55	.009	0.10	1.97	17.7	25.5	DE	0137-
40	48,000	55	.008	.090	2.08	18.7	27.1	DF	0138-

See footnotes at end of table.

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TABLE I. Style CUR91 capacitors - Continued.

DC rated voltage @ 105°C	Capacitance	DC sugre voltage @ 105°C	ESR (max) Ⓐ	Impedance (max) @ -55°C (120 Hz)	DC leakage (max) 25°C	DC leakage (max) 105°C	Maximum AC ripple current 1/ 120 Hz @ 85°C	Case size	Dash number 2/
Volts	µF	Volts	Ohms	Ohms	mA		Amps		
50	1,400	75	.077	1.94	0.40	3.57	4.6	AA	0139-
50	2,900	75	.037	0.94	0.57	5.14	7.7	AB	0140-
50	4,400	75	.027	0.64	0.70	6.33	8.8	AC	0141-
50	5,100	75	.024	0.53	0.76	6.82	9.9	AD	0142-
50	5,800	75	.021	0.47	0.81	7.27	10.9	AE	0143-
50	6,600	75	.020	0.41	0.86	7.76	11.5	AF	0144-
50	9,000	75	.018	0.30	1.00	9.05	13.0	BC	0145-
50	11,000	75	.017	0.25	1.11	10.0	15.2	BD	0146-
50	12,000	75	.016	0.23	1.16	10.5	16.4	CB	0147-
50	13,000	75	.014	0.21	1.21	10.9	16.1	BE	0148-
50	15,000	75	.012	0.18	1.30	11.7	16.8	BF	0149-
50	17,000	75	.012	0.16	1.38	12.4	19.1	CC	0150-
50	20,000	75	.010	0.14	1.50	13.5	20.4	CD	0151-
50	22,000	75	.009	0.12	1.57	14.2	21.6	CE	0152-
50	25,000	75	.009	0.11	1.68	15.1	22.4	CF	0153-
50	25,000	75	.013	0.11	1.68	15.1	20.7	DC	0154-
50	30,000	75	.011	0.10	1.84	16.5	22.5	DD	0155-
50	34,000	75	.010	.082	1.96	17.6	24.0	DE	0156-
50	39,000	75	.009	.072	2.09	18.9	26.4	DF	0157-
75	820	100	.128	3.28	0.30	2.69	4.0	AA	0158-
75	1,600	100	.068	1.68	0.41	3.70	6.6	AB	0159-
75	2,300	100	.048	1.17	0.52	4.68	7.7	AC	0160-
75	2,600	100	.044	1.03	0.55	4.96	8.6	AD	0161-
75	3,000	100	.039	0.90	0.58	5.23	9.3	AE	0162-
75	3,300	100	.035	0.81	0.64	5.73	10.3	AF	0163-
75	3,700	100	.033	0.73	0.66	5.96	10.0	BB	0164-
75	5,400	100	.021	0.50	0.81	7.30	12.9	BC	0165-
75	5,700	100	.021	0.48	0.86	7.76	13.9	CB	0166-
75	6,200	100	.020	0.44	0.87	7.84	13.4	BD	0167-
75	7,100	100	.018	0.38	0.92	8.27	14.2	BE	0168-
75	7,900	100	.017	0.30	0.99	8.90	15.1	BF	0169-
75	8,500	100	.015	0.32	1.06	9.50	16.7	CC	0170-
75	10,000	100	.013	0.27	1.14	10.3	18.6	CD	0171-
75	11,000	100	.011	0.25	1.22	11.0	19.8	CE	0172-
75	13,000	100	.010	0.21	1.29	11.6	20.6	CF	0173-
75	13,000	100	.015	0.21	1.27	11.4	19.4	DC	0174-
75	15,000	100	.013	0.18	1.36	12.3	20.7	DD	0175-
75	17,000	100	.012	0.16	1.48	13.3	22.1	DE	0176-
75	19,000	100	.010	0.14	1.54	13.8	23.7	DF	0177-
100	410	150	.200	6.58	0.24	2.13	3.5	AA	0178-
100	810	150	.105	3.33	0.34	3.02	4.8	AB	0179-
100	1,200	150	.074	2.25	0.42	3.82	6.3	AC	0180-
100	1,400	150	.065	1.93	0.45	4.05	7.3	AD	0181-
100	1,500	150	.058	1.79	0.47	4.27	7.8	AE	0182-
100	1,700	150	.053	1.59	0.50	4.48	8.2	AF	0183-

See footnotes at end of table.

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TABLE I. Style CUR91 capacitors - Continued

DC rated voltage @ 105°C	Capacitance (nom)	DC surge voltage @ 105°C	ESR (max) Ⓐ	Impedance (max) @ -55°C (120 Hz)	DC leakage (max) 25°C	DC leakage (max) 105°C	Maximum AC ripple current 1/ 120 Hz @ 85°C	Case size	Dash 2/ number
Volts	µF	Volts	Ohms	Ohms	mA		Amps		
100	1,900	150	.048	1.42	0.52	4.68	9.0	BB	0184-
100	2,800	150	.033	0.97	0.64	5.73	10.8	BC	0185-
100	2,900	150	.045	0.74	0.69	6.19	9.8	CB	0186-
100	3,200	150	.029	0.84	0.69	6.19	12.0	BD	0187-
100	3,600	150	.026	0.76	0.73	6.61	13.0	BE	0188-
100	4,100	150	.023	0.66	0.76	6.88	13.8	BF	0189-
100	4,400	150	.025	0.62	0.84	7.52	13.4	CC	0190-
100	5,100	150	.022	0.53	0.90	8.10	14.0	CD	0191-
100	5,900	150	.020	0.46	0.96	8.64	15.5	CE	0192-
100	6,500	150	.018	0.42	1.02	9.16	17.6	DC	0193-
100	6,600	150	.017	0.41	1.00	9.06	16.5	CF	0194-
100	7,600	150	.017	0.36	1.11	10.0	18.2	DD	0195-
100	8,700	150	.016	0.31	1.16	10.5	18.9	DE	0196-
100	9,500	150	.015	0.29	1.23	11.1	19.6	DF	0197-
150	210	200	.510	12.8	0.24	2.16	1.10	AA	0198-
150	380	200	.280	7.08	0.34	3.09	1.90	AB	0199-
150	550	200	.194	4.88	0.41	3.70	2.60	AC	0200-
150	640	200	.168	4.18	0.49	4.37	3.30	AE	0201-
150	720	200	.140	3.72	0.51	4.62	3.60	AF	0202-
150	900	200	.130	2.98	0.52	4.68	3.60	BB	0203-
150	1,300	200	.100	2.08	0.64	5.73	4.60	BC	0204-
150	1,400	200	.090	1.92	0.66	5.96	5.20	CB	0205-
150	1,700	200	.075	1.08	0.71	6.40	6.20	BE	0206-
150	1,900	200	.060	1.42	0.76	6.82	6.70	BF	0207-
150	2,100	200	.054	1.28	0.82	7.39	7.10	CC	0208-
150	2,800	200	.044	0.96	0.95	8.59	9.10	CE	0209-
150	3,100	200	.037	0.86	1.00	9.06	9.60	CF	0210-
150	3,100	200	.038	0.86	1.02	9.21	9.50	DC	0211-
150	4,200	200	.030	0.64	1.16	10.5	11.5	DE	0212-
150	4,700	200	.026	0.58	1.23	11.1	12.2	DF	0213-

1/ The ripple current values shown are representative values not supported by test data. See tables II and III for frequency and temperature factors.

2/ Complete dash number will include an additional letter symbol to indicate FR level (i.e., 0001M, 0026R, etc.).

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Type - Tubular.
Material - Metal.
Terminals - See figure 1.

DC rated voltage: See table I.

Reverse voltage: 1.5 volts maximum.

Capacitance (Cap.) value: See table I.

Cap. tolerance: -10, +50 percent.

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2. The manufacturer shall ... when the 100 FPM level is reached or exceeded for FPM

FR level: M, P, R, and S.

Operating temperature range: -55° to +105°C.

Burn-in: In accordance with MIL-C-39018.

DC leakage (DCL), See table I.

Equivalent series resistance (ESR): See table I.

Low temperature exposure:
DCL - See table I.
Cap. - See table I.
ESR - See table I.

Terminal strength: Method 211 of MIL-STD-202, test condition E.
Torque - 8 pound-inches for case diameter of 1.375 inch; 15 pound-inches for other case diameters.

Stability at reduced and high temperatures:
Step 1 (at +25°C):
DCL - See table I.
Cap. - See table I.
ESR - See table I.
Step 2 (at -55°C):
Impedance - See table I.
ΔCap. - Within the following percentages of the step 1 value:
5 Vdc: ±35 percent.
10 to 25 Vdc inclusive: ±25 percent.
30 to 100 Vdc inclusive: ±20 percent.
150 Vdc: ±30 percent.
Step 3 (at +25°C):
DCL - See table I.
ΔCap. - Within ±5 percent of step 1 value.
ESR - See table I.
Step 4 (at +105°C):
DCL - See table I.
ΔCap. - Within ±30 percent of step 1 value.
ESR - See table I.
Step 5 (at +25°C):
DCL - See table I.
ΔCap. - Within ±5 percent of step 1 value.
ESR - See table I.

Life: Method 108 of MIL-STD-202.
For qualification: Condition F (2,000 hours) at 85°C.
DCL - See table I.
ΔCap. - Within ±10 percent of initial measured value.
ESR - Not more than 130 percent of initial requirement.

10,000 hour (extended life and group C) at 85°C.
DCL - See table I.
ΔCap. - Within ±15 percent of initial measurement.
(A) ESR - Not more than 250 percent of initial requirement.

Case insulation: In accordance with MIL-C-39018.

Shock (specified pulse): Method 213 of MIL-STD-202, condition H.

Vibration: Low frequency - Method 201, MIL-STD-202

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition B (2 cycles, 30 minutes).
DCL - See table I
ΔCap. - Within ±5 percent of initial measured value.
ESR - See table I.
Case insulation - Shall meet initial requirements.

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Surge voltage:
DCL - See table I.
ΔCap. - Within ±6 percent of initial measured value.
ESR - See table I.

Vent: In accordance with MIL-C-39018.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - See table I.
ΔCap. - Within ±6 percent of initial measured value.
ESR - See table I.
Case insulation - Shall meet initial requirements.

High temperature exposure (500 hours at 105°C):
DCL - Not more than 200 percent of initial requirement.
ΔCap. - Within ±10 percent of initial measured value.
ESR - Not more than 115 percent of initial requirement.

AC verification:
DCL - See table I.
ΔCap. - Within ±15 percent of initial measured value.
ESR - 200 percent of initial requirement.

Reverse voltage aging: In accordance with MIL-C-39018.

Marking: In accordance with MIL-C-39018.
Part number: M39018/06-(dash number from table I).

APPLICATION NOTES:

1. Ripple current. Capacitors shall withstand the rms ripple currents at 120 Hz as given in table I.

The maximum ripple current shall apply with the following qualifications:

(A) The sum of the dc voltage plus the peak value of ac voltage must not exceed the working voltage rating of the unit, or the ac shall never exceed the maximum dc.

The values of dc voltage and ac peak voltage must be such that not more than a 1.5 volt voltage reversal is experienced.

If capacitors are used to filter circuits at a frequency other than 120 Hz, the rated 120 Hz rms ripple current should be multiplied by the factor given in the following table.

TABLE II. Factors for determining ripple currents at various frequencies.

Rated voltage	Frequency (in hertz)					
	50	60	100	120	400	1000 to 20 kHz
5-50	0.85	0.90	0.99	1.0	1.05	1.10
75-150	0.80	0.83	0.95	1.0	1.15	1.20

TABLE III. Factors for determining ripple currents at various temperatures.

Temperature (in degrees Celsius)				
25°	45°	65°	85°	105°
2.0	1.7	1.4	1.0	0.3

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2. Insulation and grounding. These capacitors have an inherently low indeterminate insulation resistance between the cathode terminal and the container. The container should be considered at the same potential as the cathode terminal with respect to ground. Where a potential difference exists between the cathode terminal and the chassis or other metallic mounting surfaces which are usually ground, the capacitors should be mounted in such a manner as to insulate the container from such mounting surfaces.

3. Shelf life. General shelf life expectancy for these capacitors is in excess of 3 years provided storage temperatures do not exceed 40°C. Storage at higher temperatures may decrease shelf life.

It is recommended that capacitors stored more than 3 years be checked at room temperature for leakage current in accordance with the applicable requirement before use.

Ⓐ 4. Torque. The minimum and maximum torque applied to the steel screw/aluminum insert shall be as follows:

	Min	Max
With a minimum of 3 engaged threads --	12 inch/lbs	18 inch/lbs
With a minimum of 6 engaged threads --	20 inch/lbs	25 inch/lbs

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-39018/6A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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Navy - EC

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